from the decay of ^{60}Co where two cascade gamma photons are emitted from the excited ^{60}Ni nucleus. These two characteristic gamma lines of 1173 keV and 1332 keV are always referred to as "cobalt gammas" whereas they are actually emitted by nickel.

5.6. SPONTANEOUS FISSION

As the nuclear charge increases to large values, nuclei become more unstable. This is reflected by decreasing half-lives for nuclei heavier than uranium. In 1940 K. Petrzak and G. Flerov found that ^{238}U in addition to α-decay also had a competing mode of radioactive decay termed *spontaneous fission*. In this mode two heavy fragments (*fission products*) are formed in addition to some neutrons. The reaction may be written

$$ {}^{A}_{Z}\mathrm{X} \rightarrow {}^{A1}_{Z1}\mathrm{X}_1 + {}^{A2}_{Z2}\mathrm{X}_2 + \nu\mathrm{n} \tag{5.52} $$

where ν is the number of neutrons, usually 2–3. The half-life for spontaneous fission of ^{238}U is very long, about 8×10^{15} y. This means that about 70 fissions occur per second in 1 kg of ^{238}U, which can be compared with the simultaneous emission of 45×10^{9} α-particles.

With increasing Z, spontaneous fission becomes more common; i.e. the half-life for this decay process decreases. For $^{240}_{94}$Pu it is 1.2×10^{11} y; for $^{244}_{96}$Cm, 1.4×10^{7} y; for $^{252}_{98}$Cf, 66 y; and for $^{256}_{100}$Fm, 3×10^{-4} y. In fact, spontaneous fission becomes the dominating decay mode for the heaviest nuclei (see Fig. 4.3).

Spontaneous fission is in some ways similar to fission induced by bombardment with low energy neutrons (§11.7).

5.7. RARE MODES OF DECAY

Radioactive decay by *proton emission* is a very seldom observed decay mode for very neutron deficient nuclides because decay by β^+ or EC normally has a very much shorter partial half-life (§5.14). Decay by p^+ has been observed for ^{53m}Co (E_p 1.55 MeV, $t_{½}$ 0.25 s, ~1.5%). However, β^+ decay sometimes leads to a proton-unstable excited state, which immediately ($<10^{-12}$ s) emits a proton. Several β^+ emitters from ^{9}C to ^{41}Ti with $N = Z - 3$ have β^+ delayed proton emission with half-lives in the range 10^{-3}–0.5 s. Also radioactive decay by simultaneous emission of two protons has been observed for a few proton rich nuclides, e.g. ^{16}Ne, $t_{½}$ ~10^{-20} s.

Among the very neutron rich nuclides, e.g. some fission products, β^- delayed neutron emission is observed. This decay mode is similar in nature to the β^+ delayed p^+ emission. Delayed n-emission is important for the safe operation of nuclear reactors, see Ch. 19.

Decay by emission of particles heavier than α, e.g. ^{12}C, ^{16}O, is energetically possible for some heavy α-emitters and has been observed in a few cases.

5.8. DECAY SCHEMES AND ISOTOPE CHARTS

Information on the mode of decay, the decay energy, and the half-life are included in the *nuclear decay scheme*. A number of simplified decay schemes are shown in Figure 5.5. Figure 4.8 explains a more detailed decay scheme, for $A = 99$.

Using Figure 4.8 as a guide, the decay schemes in Figure 5.5 are easily understood. Figure 5.5(c) shows the β^- decay of the neutron. The decay scheme of ^{137}Cs (Fig. 5.5(f)) differs somewhat from what we would expect from the curve in Figure 5.1. The reason is that in the electron spectrum of Figure 5.1, a small fraction of electrons (8%) emitted with an energy of 1.20 MeV could not be detected because of the insensitivity of the magnetic spectrometer used. It is common for nuclei to decay through different competing reactions, as in this case which involves different β-rays. If the higher energy β-decay had been as common as the lower one, we would have observed a mixed β-spectrum, as is indicated in Figure 5.6. Figure 5.5(a) shows the decay of ^{60}Co, and also its isomeric *precursor* ^{60m}Co. The β-decay is immediately followed by a cascade of two γ-rays. ^{64}Cu (Fig. 5.5(g)) decays through β^- (39%) and β^+ (17%) emission and electron capture (44%); this is referred to as *branched decay*. The vertical line in the angled arrow indicating the positron decay symbolizes the rest mass energy of the two electrons created, i.e. 1.02 MeV. Adding 1.02 MeV to 0.66 MeV gives 1.68 MeV, the Q-value for the decay from ^{64}Cu to ^{64}Ni. Figure 5.5(h) is a more complicated decay sequence for ^{238}U(α) ^{234}Th(β^-) ^{234}Pa. In the beginning of this chapter we pointed out that the decay of ^{238}U sometimes results in an excited state of the daughter ^{234}Th (in 23 out of

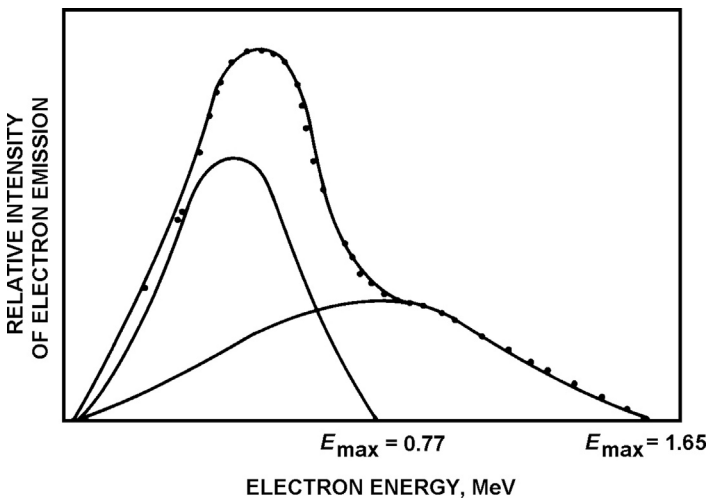


Figure 5.6 Beta spectrum for the positron decay of ^{84}Rb showing two positron groups, 0.77 MeV (11%) and 1.05 MeV (10%). (^{84}Rb also decays by β^- emission.) (Acc. to S. Eklund)